The notebeamer Package*

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1 Introduction

The notebeamer package provides an easy way to print slides on note pages quickly, developed by expl3 based on tikz. It supports various compilation methods, such as pdfLaTeX, XeLaTeX, ApLaTeX, LuaLaTeX, etc.

2 Usage

To load this package, write the line

```
\usepackage{notebeamer}
```

`\includebeamer` `\includebeamer [⟨keyvals⟩] {⟨filename⟩} [⟨keyvals⟩]`

The `\includebeamer` command can create pages with note lines, and import the specific pages of the `.pdf` file on the left side of the note pages. The mandatory argument can set the `.pdf` file that will be inserted. The optional argument accepts the following keys

`color` = ⟨*str*⟩ can set the note lines' color (Default: black), this key's name could be omitted.

`ratio` = ⟨*fp num*⟩ can set imported slides' ratio and empty area's width (Default: `0.5`).

`sep` = ⟨*dim*⟩ can set the vertical space between slides (Default: `2ex`).

`nup` = ⟨*int*⟩ can set the number of slides on every page (Default: 3).

`pages` = ⟨*comma-separated list*⟩ can select pages to insert. The list could contain ranges of page numbers. (Default: 1).

`lines` = ⟨*int*⟩ can set the number of lines on every page.

`lineno` = ⟨*true* | **false**⟩ can set whether to show the line numbers (Default: `false`).

`lhead, rhead` = ⟨*str*⟩ can set the left / right header of the pages.

*https://github.com/myhsia/notebeamer, https://ctan.org/pkg/notebeamer

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3 Working Example

Macros like \newgeometry, and \pagecolor, etc., could be used for assistance.

```
\documentclass[a4paper]{article}

\usepackage{libertine, geometry, notebeamer}

\begin{document}

\newgeometry{margin = .75in}
\pagecolor{black!80}
\includebeamer
  [ nup = 4,  pages = {2-4, 8-10},  lines = 32, ratio = .28,
    white!80, lhead = Chapter 1, lineno, rhead = Page~\thepage
  ] { example-image-a4-numbered.pdf }

\pagecolor{white}
\includebeamer
  [ nup = 5,  pages = 1-5,  lines = 32, ratio = .48,
    lhead = Chapter 1, lineno, rhead = Page~\thepage
  ] { example-image-duck.pdf }

\includebeamer
  [ nup = 2,  pages = -,  lines = 32, ratio = .54,
    lhead = Chapter 1, rhead = Page~\thepage
  ] { example-image-a4.pdf }

\end{document}
```

The effect of the above example is exhibited on the next page.

2

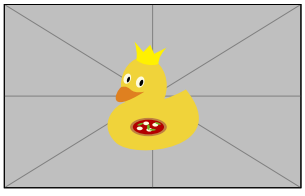
3

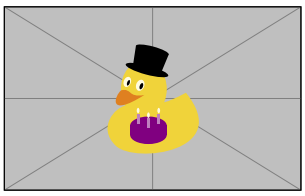
4

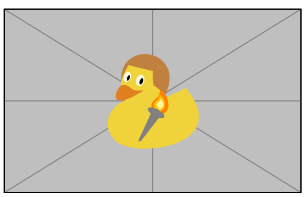
8

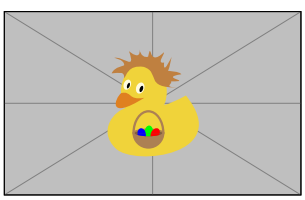
9

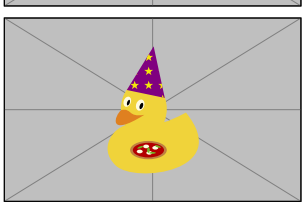
10

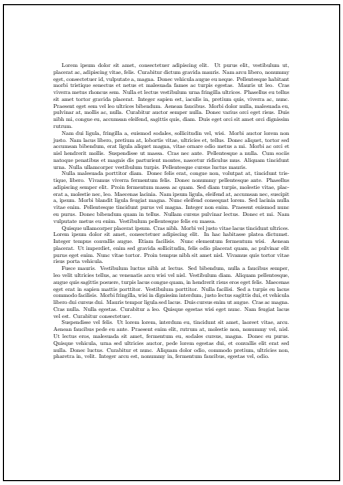












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4 The Source Code

```
⟨*package⟩
⟨@@=ntbm⟩
\def \ntbm@date    {2026-02-07}
\def \ntbm@version {v4.6A}
\ProvidesExplPackage {notebeamer} {\ntbm@date} {\ntbm@version}
  {Package for printing slides on note pages}
\cs_if_exist:NF \graphics_include:nn { \RequirePackage{l3graphics} }
\RequirePackage{tikz, tikzpagenodes}
```

4.1 User's Interface

\includebeamer

Define the \includebeamer command.

```
\NewDocumentCommand \includebeamer { O{} m O{} }
  {
    \group_begin:
    \keys_set:nn { notebeamer / includebeamer } { #1, #3 }
    \__ntbm_include_aux:n {#2}
    \group_end:
  }
```

Key–value definitions for the \includebeamer command.

\l__ntbm_include_pos_tl
\l__ntbm_include_color_tl
\l__ntbm_include_pages_tl
\l__ntbm_include_nup_int
\l__ntbm_include_lines_int
\l__ntbm_include_ratio_fp
\l__ntbm_include_sep_dim
\l__ntbm_include_lhead_tl
\l__ntbm_include_rhead_tl

```
\keys_define:nn { notebeamer / includebeamer }
  {
    pos      .tl_set:N   = \l__ntbm_include_pos_tl,
      pos    .initial:n  = left,
    color    .tl_set:N   = \l__ntbm_include_color_tl,
      color  .initial:n  = black,
    pages    .tl_set:N   = \l__ntbm_include_pages_tl,
      pages  .initial:n  = 1,
    nup      .int_set:N  = \l__ntbm_include_nup_int,
      nup    .initial:n  = 3,
    lines    .int_set:N  = \l__ntbm_include_lines_int,
    lineno   .bool_set:N = \l__ntbm_include_lineno_bool,
      lineno .initial:n = false,
      lineno .default:n = true,
    ratio    .fp_set:N   = \l__ntbm_include_ratio_fp,
      ratio  .initial:n  = .5,
    sep      .dim_set:N  = \l__ntbm_include_sep_dim,
      sep    .initial:n  = 2ex,
    lhead    .tl_set:N   = \l__ntbm_include_lhead_tl,
    rhead    .tl_set:N   = \l__ntbm_include_rhead_tl,
    unknown  .code:n     = \tl_if_novalue:nF {#1}
      { \tl_set_eq:NN \l__ntbm_include_color_tl \l_keys_key_tl }
  }
```

(*End of definition for* \includebeamer *and others. This function is documented on page* 1.)

4.2 Internal Auxiliary

\l__ntbm_nup_dim
\l__ntbm_include_ratio_dim

Store the heights and widths of the logical pages in a specific nup.

```
\dim_new:N \l__ntbm_nup_dim
\dim_new:N \l__ntbm_include_ratio_dim
```

(*End of definition for* `\l__ntbm_nup_dim` *and* `\l__ntbm_include_ratio_dim`.)

\l__ntbm_pages_total_int
\l__ntbm_pages_residue_int

Store the number of total physical pages and residue logical pages.

```
\int_new:N \l__ntbm_pages_total_int
\int_new:N \l__ntbm_pages_residue_int
```

(*End of definition for* `\l__ntbm_pages_total_int` *and* `\l__ntbm_pages_residue_int`.)

\l__ntbm_tmpa_clist

Store the results of `\__ntbm_range_to_list:nN`.

```
\clist_new:N \l__ntbm_tmpa_clist
```

(*End of definition for* `\l__ntbm_tmpa_clist`.)

__ntbm_include_aux:n

Define the auxiliary command for `\includebeamer`.

```
\cs_new_protected_nopar:Npn \__ntbm_include_aux:n #1
  {
    \graphics_get_pagecount:nN {#1} \l__ntbm_include_filepages_int
    \dim_set:Nn \l__ntbm_nup_dim { \textheight/\l__ntbm_include_nup_int }
    \dim_set:Nn \l__ntbm_include_ratio_dim
      { \fp_use:N \l__ntbm_include_ratio_fp \textwidth }
    \tl_if_eq:NnTF \l__ntbm_include_pages_tl { - }
      {
        \__ntbm_range_to_list:nN
          { 1 - \l__ntbm_include_filepages_int } \l__ntbm_tmpa_clist
      }
      {
        \exp_args:NV \__ntbm_range_to_list:nN
          { \l__ntbm_include_pages_tl } \l__ntbm_tmpa_clist
      }
    \int_set:Nn \l__ntbm_pages_total_int
      {
        \fp_eval:n
          { ceil
              ( \clist_count:N \l__ntbm_tmpa_clist/
                  \l__ntbm_include_nup_int, 0
              ) - 1
          }
      }
    \int_set:Nn \l__ntbm_pages_residue_int
      {
        \int_eval:n
          {
            \clist_count:N \l__ntbm_tmpa_clist -
            \l__ntbm_include_nup_int * \l__ntbm_pages_total_int
          }
      }
    \int_step_inline:nn { \int_use:N \l__ntbm_pages_total_int }
      {
        \clearpage
        \__ntbm_empty_note_aux:
        \int_step_inline:nn { \l__ntbm_include_nup_int }
          {
            \tikz [ remember~picture, overlay ]
              \node [ xshift = \l__ntbm_include_ratio_dim/2,
                      yshift = \fp_eval:n { -####1 + .5 } \l__ntbm_nup_dim
                    ] at (current~page~text~area.north~west)
```

```
                {
                  \includegraphics
                    [ height = \dim_eval:n
                        { \l__ntbm_nup_dim - \l__ntbm_include_sep_dim },
                      page = \clist_item:Nn \l__ntbm_tmpa_clist
                        { ####1 + \l__ntbm_include_nup_int * ( ##1 - 1 ) }
                    ] {#1}
                };
            }
          \clearpage
        }
      \__ntbm_empty_note_aux:
      \int_step_inline:nn { \int_use:N \l__ntbm_pages_residue_int }
        {
          \tikz [ remember~picture, overlay ]
            \node [ xshift = \l__ntbm_include_ratio_dim/2,
                    yshift = \fp_eval:n {( -##1 + .5 )} \l__ntbm_nup_dim
                  ] at (current~page~text~area.north~west)
              {
                \includegraphics
                  [ height = \dim_eval:n
                      { \l__ntbm_nup_dim - \l__ntbm_include_sep_dim },
                    page = \clist_item:Nn \l__ntbm_tmpa_clist
                      {
                        \l__ntbm_include_nup_int *
                        \l__ntbm_pages_total_int + ##1
                      }
                  ] {#1}
              };
        }
      \clearpage
    }
```

(*End of definition for* `\__ntbm_include_aux:n`.)

`\__ntbm_empty_note_aux:` Define the auxiliary command for creating empty note line page.

```
\cs_new_protected_nopar:Nn \__ntbm_empty_note_aux:
  {
    \begin { tikzpicture } [ remember~picture, overlay ]
      \draw [ very~thick, \l__ntbm_include_color_tl, opacity = .8 ]
        (current~page~text~area.north~west) --
        (current~page~text~area.north~east)
       node [ at~start, above~right, font = \Large \bfseries ]
        { \l__ntbm_include_lhead_tl }
       node [ above~left, font = \Large \bfseries ]
        { \l__ntbm_include_rhead_tl };
      \draw [ very~thick, \l__ntbm_include_color_tl, opacity = .8 ]
        (current~page~text~area.south~west) --
        (current~page~text~area.south~east);
      \int_step_inline:nn { \l__ntbm_include_lines_int - 1 }
        {
          \draw [ thick, \l__ntbm_include_color_tl, opacity = .6 ]
            ([xshift = \l__ntbm_include_ratio_dim,
              yshift = -\textheight/\l__ntbm_include_lines_int * ##1
```

```
            ]current~page~text~area.north~west) --++
            (\dim_eval:n { \textwidth - \l__ntbm_include_ratio_dim },0);
          \bool_if:nT \l__ntbm_include_lineno_bool
            {
              \node [ font = \footnotesize, \l__ntbm_include_color_tl,
                      opacity = .4, right ] at
                ([xshift = \textwidth,
                  yshift = -\textheight/\l__ntbm_include_lines_int * (##1 - .5)
                 ]current~page~text~area.north~west)
                { \int_eval:n { ##1 } };
            }
        }
      \node [ font = \footnotesize, \l__ntbm_include_color_tl,
              opacity = .4, right ] at
        ([xshift = \textwidth,
          yshift = -\textheight + \textheight/\l__ntbm_include_lines_int * .5
         ]current~page~text~area.north~west)
        { \int_eval:n { \l__ntbm_include_lines_int } };
    \end { tikzpicture }
    \pagestyle { empty }
  }
```

(*End of definition for* `\__ntbm_empty_note_aux:`.)

`\l__ntbm_tmpa_seq` `\l__ntbm_tmpb_seq` Store the results of 2D array segmentation.

```
\seq_new:N \l__ntbm_tmpa_seq
\seq_new:N \l__ntbm_tmpb_seq
```

(*End of definition for* `\l__ntbm_tmpa_seq` *and* `\l__ntbm_tmpb_seq`.)

`\__ntbm_range_to_list:nN` Convert the combination of number and number range to a list.

```
\cs_new_protected_nopar:Npn \__ntbm_range_to_list:nN #1#2
  {
    \clist_clear:N #2
    \seq_set_split:Nnn \l__ntbm_tmpa_seq { , } {#1}
    \seq_map_inline:Nn \l__ntbm_tmpa_seq
      {
        \tl_if_in:nnTF {##1} { - }
          {
            \seq_set_split:Nnn \l__ntbm_tmpb_seq { - } {##1}
            \int_step_inline:nnn
              { \seq_item:Nn \l__ntbm_tmpb_seq { 1 } }
              { \seq_item:Nn \l__ntbm_tmpb_seq { 2 } }
              { \clist_put_right:Nn #2 {####1} }
          } { \clist_put_right:Nn #2 {##1} }
      }
  }
```

(*End of definition for* `\__ntbm_range_to_list:nN`.)

```
⟨/package⟩
```

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